INDIVIDUAL REFLECTION

Mini Use Case 02 – Credit Pre-Screening

Calibo AI Academy | Part 2 – KLU Batch | Phase 2 | Supervised Learning Part 1

Team Member	Medpati Akhilendra Reddy
Role	ML Analyst – Fairness & Credit Policy Review
Contribution	Step 5.6: Fairness Check and Step 5.7: Credit Policy Briefing

1. Project Overview

Our team worked on Mini Use Case 02 in the BFSI – Retail Lending (Personal Loans) domain. The objective was to examine historical loan decisions and develop a credit pre-screening approach using supervised learning. My contribution focused on checking fairness in model outcomes and translating the model evaluation into practical credit-policy recommendations. The model predicts historical analyst decisions; its predictions should not be interpreted as direct evidence of future repayment or default.

2. My Role and Contribution

Step 5.6 — Fairness Check. I compared historical approval patterns and model auto-decline outcomes across gender and employment groups. Historical approval was 54.1% for male applicants and 50.4% for female applicants, a 3.7 percentage-point difference. I also reviewed income and employment composition to understand potential proxy effects: median income was ■55,000 for male applicants and ■45,000 for female applicants, while gig/contract applicants represented 11.6% of the male group and 21.4% of the female group. Employment-group approval gaps ranged from −0.5 percentage points for business applicants to +4.4 points for self-employed applicants.

At the proposed 90% decline-probability threshold, the auto-decline rate was 15.0% for male applicants and 15.6% for female applicants. Among auto-declined applicants, the historical-approval (wrongful-decline) share was 5.2% for males (13 of 248) and 3.4% for females (4 of 117). I treated these results as monitoring signals rather than proof of fairness, because group differences can reflect applicant mix and historical decision patterns.

Step 5.7 — Credit Policy Briefing. I translated the threshold and fairness results into policy guidance. The 90% decline-probability threshold was proposed for a controlled pilot rather than immediate full deployment. At this threshold, the estimated share of historically approved applicants among auto-declines was about 4.66%, compared with approximately 20.8% at a 0.50 threshold. For an estimated 3,000 applications per month, the proposed threshold could remove about 456 cases from the manual queue and free approximately 190 analyst hours, assuming 25 minutes per application. These are estimates and should be validated against actual workflow data.

3. Key Findings

Measure	Result	Interpretation
Historical approval: male / female	54.1% / 50.4%	3.7 pp historical approval gap; investigate context and group composition.
Median income: male / female	■55,000 / ■45,000	Income differences may affect outcomes and act as proxy signals.
Gig/contract share: male / female	11.6% / 21.4%	Different employment mix may influence group comparisons.
Auto-decline rate at 90%: male / female	15.0% / 15.6%	Rates are close, but should be monitored over time.
Wrongful-decline share: male / female	5.2% (13/248) / 3.4% (4/117)	Review declined cases and validate before operational use.
90% threshold: estimated monthly impact	456 cases; ~190 analyst hours	Potential workload benefit; confirm using real operating data.

4. Interpretation and Recommendation

The fairness review indicates that historical approval and model auto-decline patterns should be checked across groups, with attention to income and employment variables that may act as proxies. Gender should not be used as a model decision feature. I recommend a controlled pilot using the 90% threshold, human review for new-to-credit (NTC)

applicants rather than automatic rejection, and monthly monitoring of approval rates, auto-decline rates, and wrongful-decline rates by gender and employment group. Before broader deployment, the model should be tested on out-of-time data and evaluated against actual repayment/default outcomes. A model that reproduces historical decisions is not automatically a model of creditworthiness.

5. Technical Work and Learning

My work involved interpreting model threshold outputs, comparing group-level rates, calculating percentage-point differences, and connecting fairness metrics with a practical operating policy. I learned that fairness cannot be judged from one metric alone: approval rates, auto-decline rates, wrongful declines, and applicant composition should be considered together. I also learned to distinguish model confidence from certainty and to treat workload savings as estimates that require operational validation.

6. Final Summary

My contribution covered Steps 5.6 and 5.7 of Mini Use Case 02. I reviewed gender- and employment-related fairness patterns, assessed auto-decline outcomes at a proposed 90% threshold, and converted the findings into a cautious credit-policy briefing. The analysis highlighted the need for transparent monitoring, human review for new-to-credit applicants, and validation before deployment. The proposed threshold may reduce manual workload, but responsible implementation requires ongoing fairness checks and evaluation against actual repayment outcomes.

Individual Reflection: This work helped me understand how machine-learning outputs can influence real financial decisions. I gained practical experience in fairness evaluation, threshold analysis, and communicating model limitations to support responsible decision-making.